

10th Grade Midterm Analysis of Decisions: C4 Solutions

Evaluated Criterion

C7: Uses subjective probabilities and sensitivity analysis to examine how changes in probability assumptions affect the preferred decision.

1 Expected Value and Sensitivity Analysis

When probabilities are subjective or estimated, a decision-maker can examine how changes in the probability assumptions affect the preferred alternative. The Expected Value criterion allows each alternative to be represented as a function of the probability assigned to the possible states of nature. For a payoff table with alternatives in rows and states of nature in columns, let $P_{ay}(A_i, S_j)$ represent the payoff obtained when alternative A_i is selected and state of nature S_j occurs.

Let p represent the subjective probability that state of nature S_1 occurs. Since there are two possible states,

$$P(S_1) = p \quad \text{and} \quad P(S_2) = 1 - p \quad \text{with} \quad 0 \leq p \leq 1$$

For each alternative, calculate its Expected Value as a function of p :

$$EV(A_i) = pP_{ay}(A_i, S_1) + (1 - p)P_{ay}(A_i, S_2).$$

This expression shows how the expected payoff changes when the subjective probability of S_1 changes. The preferred alternative depends on the value of p . To determine which alternative is preferred, compare their expected-value functions. For two alternatives A_1 and A_2 , the threshold probability is found by solving

$$EV(A_1) = EV(A_2).$$

At this threshold, the decision-maker is indifferent between the two alternatives. The preferred alternative on each side of the threshold is determined by comparing the corresponding expected values. Therefore, sensitivity analysis identifies how changes in the subjective probability affect the preferred decision.

2 General 2x2 Problem

Consider the following payoff table, where the probability of state S_1 is p and the probability of state S_2 is $1 - p$:

Alternative	S_1	S_2
	(p)	$(1 - p)$
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$

Determine the preferred alternative as a function of p , including the threshold probability at which the decision-maker is indifferent.

2.1 Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1, A_2 States of nature: S_1, S_2

Probabilities: $P(S_1) = p$ $P(S_2) = 1 - p$

Step 2: Calculate the expected value of A_1

For alternative A_1 :

$$EV(A_1) = pP_{ay}(A_1, S_1) + (1 - p)P_{ay}(A_1, S_2).$$

This expression shows how the expected payoff of A_1 changes as p changes.

Step 3: Calculate the expected value of A_2

For alternative A_2 :

$$EV(A_2) = pP_{ay}(A_2, S_1) + (1 - p)P_{ay}(A_2, S_2).$$

This expression shows how the expected payoff of A_2 changes as p changes.

Step 4: Find the threshold probability

The threshold occurs when the two alternatives have the same expected value:

$$EV(A_1) = EV(A_2).$$

Solve this equation for p . The resulting value is the probability at which the decision-maker is indifferent between the two alternatives.

Step 5: Perform the sensitivity analysis

Compare the expected values on each side of the threshold.

Decision rule

$$\begin{array}{lll} EV(A_1) > EV(A_2) & \Rightarrow & \text{choose } A_1, \\ EV(A_2) > EV(A_1) & \Rightarrow & \text{choose } A_2, \\ EV(A_1) = EV(A_2) & \Rightarrow & \text{indifferent.} \end{array} \quad \longrightarrow \quad \left\{ \begin{array}{ll} A_1, & \text{if } EV(A_1) > EV(A_2), \\ A_2, & \text{if } EV(A_2) > EV(A_1), \\ \text{indifferent,} & \text{if } EV(A_1) = EV(A_2). \end{array} \right.$$

Thus, the preferred alternative can be expressed as a function of the subjective probability p .

Extended payoff table

Alternative	S_1 (p)	S_2 ($1 - p$)	Expected Value
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$pP_{ay}(A_1, S_1) + (1 - p)P_{ay}(A_1, S_2)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$pP_{ay}(A_2, S_1) + (1 - p)P_{ay}(A_2, S_2)$

3 General 4x3 Problem

Consider the following payoff table, where the probabilities of the states of nature are subjective and may vary:

Alternative	S_1 (p)	S_2 (q)	S_3 ($1 - p - q$)	$P(S_1) = p,$	$P(S_2) = q,$	$P(S_3) = 1 - p - q$
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$P_{ay}(A_1, S_3)$	$p \geq 0,$	$q \geq 0,$	$p + q \leq 1$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$P_{ay}(A_2, S_3)$			
A_3	$P_{ay}(A_3, S_1)$	$P_{ay}(A_3, S_2)$	$P_{ay}(A_3, S_3)$			
A_4	$P_{ay}(A_4, S_1)$	$P_{ay}(A_4, S_2)$	$P_{ay}(A_4, S_3)$			

Determine the preferred alternative as a function of the subjective probabilities p and q , including any threshold values or boundaries at which the preferred decision changes.

3.1 Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1, A_2, A_3, A_4 States of nature: S_1, S_2, S_3

Probabilities: $P(S_1) = p,$ $P(S_2) = q,$ $P(S_3) = 1 - p - q.$

Step 2: Calculate the expected value of A_1

$$EV(A_1) = pP_{ay}(A_1, S_1) + qP_{ay}(A_1, S_2) + (1 - p - q)P_{ay}(A_1, S_3).$$

This expression shows how the expected payoff of A_1 changes as the subjective probabilities change.

Step 3: Calculate the expected value of A_2

$$EV(A_2) = pP_{ay}(A_2, S_1) + qP_{ay}(A_2, S_2) + (1 - p - q)P_{ay}(A_2, S_3).$$

Step 4: Calculate the expected value of A_3

$$EV(A_3) = pP_{ay}(A_3, S_1) + qP_{ay}(A_3, S_2) + (1 - p - q)P_{ay}(A_3, S_3).$$

Step 5: Calculate the expected value of A_4

$$EV(A_4) = pP_{ay}(A_4, S_1) + qP_{ay}(A_4, S_2) + (1 - p - q)P_{ay}(A_4, S_3).$$

Step 6: Compare the expected-value functions and perform the sensitivity analysis

Compare

$$EV(A_1), \quad EV(A_2), \quad EV(A_3), \quad EV(A_4).$$

The preferred alternative depends on the values of p and q . Examine the preferred alternative for different feasible combinations of p and q , subject to

$$p \geq 0, \quad q \geq 0, \quad p + q \leq 1.$$

For any two alternatives, a boundary occurs when their expected values are equal, indicating indifference. The preferred alternative is the one with the highest expected value:

$$EV(A_i) = EV(A_k) \quad \text{at a boundary,} \quad A^* = \arg \max_i EV(A_i)$$

Extended payoff table

Alternative	S_1 p	S_2 q	S_3 $1 - p - q$	Expected Value
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$P_{ay}(A_1, S_3)$	$pP_{ay}(A_1, S_1) + qP_{ay}(A_1, S_2) + (1 - p - q)P_{ay}(A_1, S_3)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$P_{ay}(A_2, S_3)$	$pP_{ay}(A_2, S_1) + qP_{ay}(A_2, S_2) + (1 - p - q)P_{ay}(A_2, S_3)$
A_3	$P_{ay}(A_3, S_1)$	$P_{ay}(A_3, S_2)$	$P_{ay}(A_3, S_3)$	$pP_{ay}(A_3, S_1) + qP_{ay}(A_3, S_2) + (1 - p - q)P_{ay}(A_3, S_3)$
A_4	$P_{ay}(A_4, S_1)$	$P_{ay}(A_4, S_2)$	$P_{ay}(A_4, S_3)$	$pP_{ay}(A_4, S_1) + qP_{ay}(A_4, S_2) + (1 - p - q)P_{ay}(A_4, S_3)$

4 Contextualized 2x2 Problem - Coffee shop

A small coffee shop is considering whether to introduce a new dessert line or keep its current menu. The manager is uncertain about the level of customer traffic during the promotional weekend. Let p represent the subjective probability of high customer traffic. Therefore, the probability of low customer traffic is $1 - p$. All payoffs are expressed in thousands of dollars. Using the Expected Value criterion, determine how the preferred decision changes as the subjective probability p changes. Identify the threshold probability at which the manager is indifferent between the two alternatives.

Alternative	High traffic S_1	Low traffic S_2
	$P(S_1) = p$	$P(S_2) = 1 - p$
Launch dessert line (A_1)	40	10
Keep current menu (A_2)	30	20

4.1 Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Launch dessert line, A_2 = Keep current menu

States of nature: S_1 = High traffic, S_2 = Low traffic

Probabilities: $P(S_1) = p, P(S_2) = 1 - p$ with $0 \leq p \leq 1$.

Step 2: Calculate the expected value of A_1

For the Launch dessert line alternative:

$$\begin{aligned} EV(A_1) &= p(40) + (1 - p)(10) \\ &= 40p + 10 - 10p = 30p + 10. \end{aligned}$$

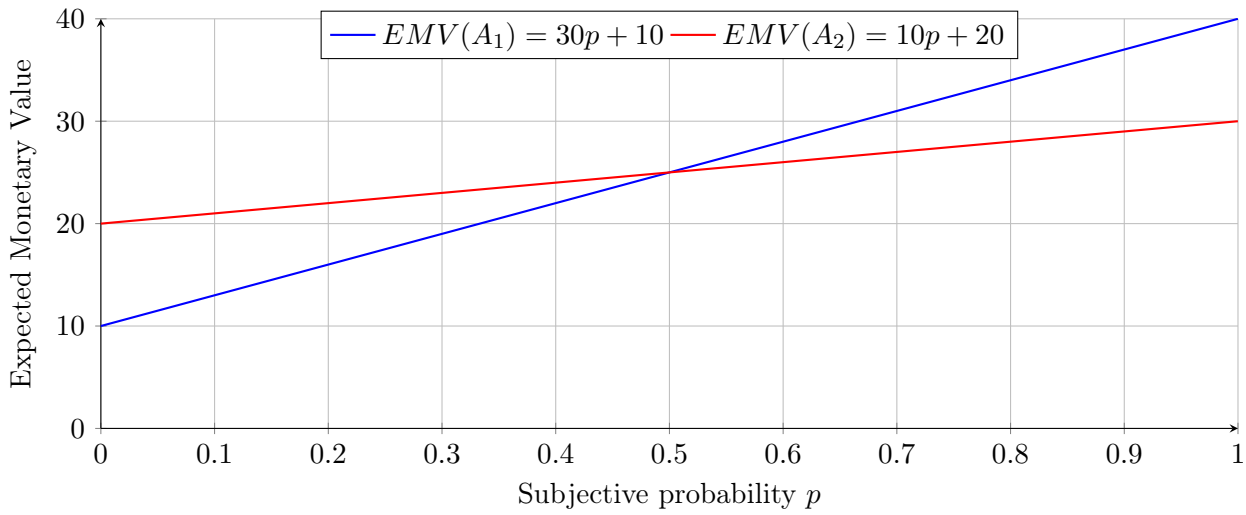
Thus, the expected payoff from launching the dessert line is $30p + 10$ thousand dollars.

Step 3: Calculate the expected value of A_2

For the Keep current menu alternative:

$$\begin{aligned} EV(A_2) &= p(30) + (1 - p)(20) \\ &= 30p + 20 - 20p = 10p + 20. \end{aligned}$$

Thus, the expected payoff from keeping the current menu is $10p + 20$ thousand dollars.



Step 4: Find the threshold probability

The threshold occurs when the two alternatives have the same expected value:

$$\begin{aligned}
 EV(A_1) &= EV(A_2) &\longrightarrow& 30p + 10 = 10p + 20 \\
 \longrightarrow & 30p - 10p = 20 - 10 &\longrightarrow& 20p = 10 &\longrightarrow& p = \frac{10}{20} = \frac{1}{2} = 0.5
 \end{aligned}$$

Thus, the manager is indifferent between the two alternatives when the subjective probability of high customer traffic is 50%.

Step 5: Perform the sensitivity analysis

Compare the expected values on each side of the threshold.

$$\begin{aligned}
 p < 0.5 &\Rightarrow EV(A_1) < EV(A_2) \Rightarrow \text{choose } A_2, \\
 p > 0.5 &\Rightarrow EV(A_1) > EV(A_2) \Rightarrow \text{choose } A_1, \\
 p = 0.5 &\Rightarrow EV(A_1) = EV(A_2) \Rightarrow \text{indifferent.}
 \end{aligned}
 \longrightarrow
 \begin{cases}
 A_2, & 0 \leq p < 0.5, \\
 \text{indifferent,} & p = 0.5, \\
 A_1, & 0.5 < p \leq 1.
 \end{cases}$$

The sensitivity analysis shows that the coffee shop should keep its current menu when the subjective probability of high traffic is below 50%. When that probability exceeds 50%, launching the dessert line has the higher expected value.

Extended payoff table

Alternative	High traffic S_1	Low traffic S_2	Expected Value
	$P(S_1) = p$	$P(S_2) = 1 - p$	
A_1 : Launch dessert line	40	10	$p(40) + (1 - p)(10) = 30p + 10$
A_2 : Keep current menu	30	20	$p(30) + (1 - p)(20) = 10p + 20$

5 Contextualized 2x2 Problem - Clothing store

A local clothing store is deciding whether to launch a new advertising campaign before the holiday season or continue with its current marketing strategy. The manager is uncertain about the level of sales during the holiday season. Let p represent the subjective probability of strong sales. Therefore, the probability of weak sales is $1 - p$. All payoffs are expressed in thousands of dollars.

Using the Expected Value criterion, determine how the preferred decision changes as the subjective probability p changes. Identify the threshold probability at which the manager is indifferent between the two alternatives.

Alternative	Strong sales S_1	Weak sales S_2
	$P(S_1) = p$	$P(S_2) = 1 - p$
Launch new campaign (A_1)	55	5
Keep current strategy (A_2)	45	45

5.1 Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Launch new campaign, A_2 = Keep current strategy

States of nature: S_1 = Strong sales, S_2 = Weak sales

Probabilities: $P(S_1) = p, P(S_2) = 1 - p$ with $0 \leq p \leq 1$.

Step 2: Calculate the expected value of A_1

For the Launch new campaign alternative:

$$\begin{aligned}EV(A_1) &= p(55) + (1 - p)(5) \\&= 55p + 5 - 5p = 50p + 5.\end{aligned}$$

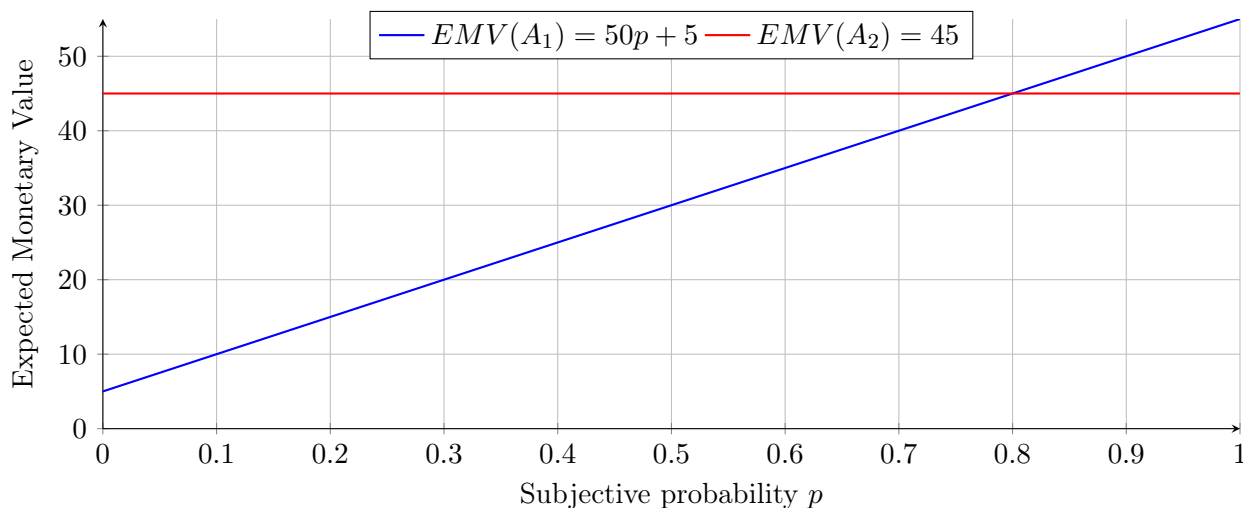
Thus, the expected payoff from launching the new campaign is $50p + 5$ thousand dollars.

Step 3: Calculate the expected value of A_2

For the Keep current strategy alternative:

$$\begin{aligned}EV(A_2) &= p(45) + (1 - p)(45) \\&= 45p + 45 - 45p = 45.\end{aligned}$$

Thus, the expected payoff from keeping the current strategy is 45 thousand dollars.



Step 4: Find the threshold probability

The threshold occurs when the two alternatives have the same expected value:

$$\begin{aligned}EV(A_1) = EV(A_2) &\longrightarrow 50p + 5 = 45 \\ \longrightarrow 50p = 40 &\longrightarrow p = 0.8.\end{aligned}$$

Thus, the manager is indifferent between the two alternatives when the subjective probability of strong sales is 80%.

Step 5: Perform the sensitivity analysis

Compare the expected values on each side of the threshold.

$$\begin{aligned}p < 0.800 &\Rightarrow EV(A_1) < EV(A_2) \Rightarrow \text{choose } A_2, \\ p > 0.800 &\Rightarrow EV(A_1) > EV(A_2) \Rightarrow \text{choose } A_1, \\ p = 0.800 &\Rightarrow EV(A_1) = EV(A_2) \Rightarrow \text{indifferent.}\end{aligned} \longrightarrow \begin{cases} A_2, & 0 \leq p < 0.800, \\ \text{indifferent,} & p = 0.800, \\ A_1, & 0.800 < p \leq 1. \end{cases}$$

The sensitivity analysis shows that the clothing store should keep its current marketing strategy when the subjective probability of strong sales is below 80%. Only when that probability exceeds 80% does launching the new advertising campaign have the higher expected value.

Extended payoff table

Alternative	Strong sales S_1	Weak sales S_2	Expected Value
	$P(S_1) = p$	$P(S_2) = 1 - p$	
A_1 : Launch new campaign	55	5	$p(55) + (1 - p)(5) = 50p + 5$
A_2 : Keep current strategy	45	45	$p(45) + (1 - p)(45) = 45$

6 Contextualized 3x2 Problem - Technology company

A technology company is deciding which product strategy to adopt for the launch of a new device. The manager is uncertain about the level of demand during the launch period. Let p represent the subjective probability of high demand. Therefore, the probability of low demand is $1 - p$. All payoffs are expressed in thousands of dollars.

Using the Expected Value criterion, determine how the preferred decision changes as the subjective probability p changes. Identify the threshold probabilities at which the manager is indifferent between the alternatives.

Alternative	High demand S_1	Low demand S_2
	$P(S_1) = p$	$P(S_2) = 1 - p$
Premium version (A_1)	80	-20
Standard version (A_2)	60	18
Budget version (A_3)	40	25

6.1 Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Premium version, A_2 = Standard version, A_3 = Budget version

States of nature: S_1 = High demand, S_2 = Low demand

Probabilities: $P(S_1) = p, P(S_2) = 1 - p$ with $0 \leq p \leq 1$.

Step 2: Calculate the expected value of A_1

For the Premium version alternative:

$$\begin{aligned}EV(A_1) &= p(80) + (1 - p)(-20) \\&= 80p - 20 + 20p \\&= 100p - 20.\end{aligned}$$

Thus, the expected payoff from the Premium version is $100p - 20$ thousand dollars.

Step 4: Calculate the expected value of A_3

For the Budget version alternative:

$$\begin{aligned}EV(A_3) &= p(40) + (1 - p)(25) \\&= 40p + 25 - 25p \\&= 15p + 25.\end{aligned}$$

Thus, the expected payoff from the Budget version is $15p + 25$ thousand dollars.

Step 6: Find the second threshold probability

For A_1 and A_2 :

$$\begin{aligned}EV(A_1) &= EV(A_2) \\100p - 20 &= 42p + 18 \\58p &= 38 \\p &= \frac{38}{58} = \frac{19}{29} \approx 0.655.\end{aligned}$$

Thus, the manager is indifferent between A_2 and A_3 when the subjective probability of high demand is approximately 25.9%, and indifferent between A_1 and A_2 when it is approximately 65.5%.

Step 7: Perform the sensitivity analysis

Compare the expected values on each side of the thresholds.

Step 3: Calculate the expected value of A_2

For the Standard version alternative:

$$\begin{aligned}EV(A_2) &= p(60) + (1 - p)(18) \\&= 60p + 18 - 18p \\&= 42p + 18.\end{aligned}$$

Thus, the expected payoff from the Standard version is $42p + 18$ thousand dollars.

Step 5: Find the threshold probabilities

The thresholds occur when two alternatives have the same expected value.

For A_2 and A_3 :

$$\begin{aligned}EV(A_2) &= EV(A_3) \\42p + 18 &= 15p + 25 \\27p &= 7 \\p &= \frac{7}{27} \approx 0.259.\end{aligned}$$

$$\begin{aligned}
p < 0.259 &\Rightarrow EV(A_3) > EV(A_2), EV(A_1) \Rightarrow \text{choose } A_3, \\
p = 0.259 &\Rightarrow EV(A_2) = EV(A_3) \Rightarrow \text{indifferent between } A_2 \text{ and } A_3, \\
0.259 < p < 0.655 &\Rightarrow EV(A_2) > EV(A_1), EV(A_3) \Rightarrow \text{choose } A_2, \\
p = 0.655 &\Rightarrow EV(A_1) = EV(A_2) \Rightarrow \text{indifferent between } A_1 \text{ and } A_2, \\
p > 0.655 &\Rightarrow EV(A_1) > EV(A_2), EV(A_3) \Rightarrow \text{choose } A_1.
\end{aligned}$$

Therefore, the preferred decision depends on the subjective probability of high demand:

$$\left\{ \begin{array}{ll} A_3, & 0 \leq p < 0.259, \\ \text{indifferent between } A_2 \text{ and } A_3, & p = 0.259, \\ A_2, & 0.259 < p < 0.655, \\ \text{indifferent between } A_1 \text{ and } A_2, & p = 0.655, \\ A_1, & 0.655 < p \leq 1. \end{array} \right.$$

The sensitivity analysis shows that the company should choose the Budget version when the subjective probability of high demand is below approximately 25.9%. Between 25.9% and 65.5%, the Standard version has the highest expected value. When the probability of high demand exceeds 65.5%, the Premium version becomes the preferred decision.

Extended payoff table

Alternative	High demand S_1	Low demand S_2	Expected Value
	$P(S_1) = p$	$P(S_2) = 1 - p$	
A_1 : Premium version	80	-20	$p(80) + (1 - p)(-20)$ $= 100p - 20$
A_2 : Standard version	60	18	$p(60) + (1 - p)(18)$ $= 42p + 18$
A_3 : Budget version	40	25	$p(40) + (1 - p)(25)$ $= 15p + 25$